Region-Adaptive CNN-Transformer Ensembles for Brain Tumour Segmentation with External Validation and Disagreement-Based Quality Control

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ABSTRACT

Convolutional and transformer networks for brain tumour segmentation make different errors, but ensembles usually average them with one weight for every tumour region and are rarely tested outside their training population. The contribution to artificial intelligence is a region-adaptive hybrid ensemble of nnU-Net, SegResNet and Swin UNETR-v2 that selects one convex weight vector and one decision threshold per tumour region on validation cases, plus a quality-control signal from the disagreement between its members. The engineering application is a segmentation service that must flag unreliable cases for review. All networks were trained at native resolution on 257 Brain Tumour Segmentation (BraTS) 2020 cases, selected on 56, and tested on 56 internal and 60 external BraTS-Africa cases. The ensemble reached a mean Dice of 0.8530 (95% confidence interval 0.8204–0.8836) internally and 0.7966 externally. It beat nnU-Net and Swin UNETR internally (Holm-corrected Wilcoxon $p < 0.01$) but not SegResNet alone, which scored 0.8717 internally ($p = 0.018$ in its favour) and 0.7967 externally. Every model lost 0.047 to 0.075 Dice externally, mostly in the tumour core. Disagreement between the members detected cases with mean Dice below 0.80 with an area under the receiver operating characteristic curve of 0.993 internally and 0.870 externally, against 0.716 and 0.791 for test-time-augmentation spread. Seg-Grad-CAM attribution was 4 to 23 times enriched in the tumour for the convolutional networks but near chance for Swin UNETR. Dropping sequences confirmed reliance on FLAIR for the whole tumour and on T1CE for the core. The disagreement between architectures proved more useful than their fusion.

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